

HW04

Sam Ly

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Exercise 1 [2]

Tools and resources used: Mainly Google, and Gemini when I got really stuck.

Exercise 1 **Points attempted:** 2. **Time spent:** 10 minutes. **Difficulty:** 1/10. **Self-assessed score:** 2/2. **Questions/Feedback:** None.

Exercise 2 **Points attempted:** 7. **Time spent:** 120 minutes. **Difficulty:** 7/10. **Self-assessed score:** 7/7. **Questions/Feedback:** Correctness

Exercise 3 **Points attempted:** 1. **Time spent:** 5 minutes. **Difficulty:** 1/10. **Self-assessed score:** 1/1. **Questions/Feedback:** Correctness

Choice Exercise 6 **Points attempted:** 10. **Time spent:** 5 minutes. **Difficulty:** 8/10. **Self-assessed score:** 10/10. **Questions/Feedback:** Correctness

Exercise 2 [7]

1. We defined the dihedral group in terms of its group presentation:

$$D_n = \langle r, f \mid r^n = f^2 = (rf)^2 = e \rangle.$$

- (a) Prove that $r^k f = f r^{n-k}$ for all $0 < k < n$.

Proof. First, we notice that $rfrf = e$. By right multiplying by f we get $rfr = f$. By left multiplying by r^{-1} , we get $fr = r^{-1}f$. Then, we left and right multiply by f on both sides to get $rf = fr^{-1}f$.

In other words, $r^1 f = f r^{n-1}$. Now, to prove that $r^k f = f r^{n-k}$ for all $0 < k < n$, we must use induction. We have already proved the base case $k = 1$. As our inductive hypothesis, we assume that the proposition is true for some value k . Thus, $r^k f = f r^{n-k}$. Now, for by left multiplying by r , we get

$$r^{k+1} f = r f r^{n-k}.$$

Now, we apply $rf = f r^{n-1}$ to the RHS, and get

$$r^{k+1} f = f r^{-1} r^{n-k} = f r^{n-(k+1)}.$$

Therefore, $r^k f = f r^{n-k}$ for all $0 < k < n$. □

- (b) Prove that every element can be written in the form $r^k f^\ell$ for $0 \leq k < n$ and $\ell \in \{0, 1\}$

Proof. We approach this problem from a formal languages angle. We begin by constructing all possible ‘words’ of the language formed by the alphabet $\{r, s\}$. One example of such a word is $rrrfrfffrfrrrrrrrf$. By definition, this language must encompass all of D_n for any n .

Then, we can form some basic substitution rules, in other words, a ‘grammar’ that allows us to rewrite the word into a nicer form. By definition of the dihedral group, we have the rules:

$$\underbrace{rr \dots r}_{m \text{ terms}} = r^m = r^{m-n} \tag{1}$$

$$ff = e. \tag{2}$$

Thus, we can rewrite our word into the form $r^{k_0} f^{l_0} r^{k_1} f^{l_1} \dots$, where $0 \leq k_i < n$ and $l_i \in \{0, 1\}$.

Now, from here, we apply the proposition proved above to combine and move all l_i to the right of the expression.

For example, if $l_0 = 0$, then the expression becomes $r^{k_0} r^{k_1} f^{l_1} \dots = r^{k_0+k_1} f^{l_1} \dots$. If $l_0 = 1$, then the expression becomes

$$r^{k_0} (f r^{k_1}) f^{l_1} \dots = r^{k_0} (r^{n-k_1} f) f^{l_1} \dots$$

which we can simplify by combining $f f^{l_1}$. This process can be repeated until we reach the form $r^k f^\ell$ for $0 \leq k < n$ and $\ell \in \{0, 1\}$. □

- (c) Prove that the order (number of elements) of the group is $D_n = 2n$.

Proof. Now that we have established that all elements of D_n can be written as the form $r^k f^\ell$ for $0 \leq k < n$ and $\ell \in \{0, 1\}$, this question can be reframed as asking “How many unique ways can we write $r^k f^\ell$?” By combinatorics, we see that the number of unique ways is $2n$. □

2. We say that a group $(G, *)$ is an abelian group if and only if $a * b = b * a$ for all $a, b \in G$.

- (a) Prove that if G is generated by g_1 and g_2 then G is abelian if and only if $g_1 * g_2 = g_2 * g_1$.

Proof. First, we prove that if $a * b = b * a$ for all $a, b \in G$, then $g_1 * g_2 = g_2 * g_1$. By setting $a = g_1$ and $b = g_2$, then $g_1 * g_2 = g_2 * g_1$.

The converse “if $g_1 * g_2 = g_2 * g_1$, then $a * b = b * a$ for all $a, b \in G$ ” is much more difficult to prove. Notice that

$$a = a_1 * a_2 * \cdots * a_p = \prod_{i=1}^p a_i \quad (3)$$

$$b = b_1 * b_2 * \cdots * b_q = \prod_{i=1}^q b_i \quad (4)$$

$$c = a * b = a_1 * a_2 * \cdots * a_p * b_1 * b_2 * \cdots * b_q \quad (5)$$

where $a_i, b_i \in \{g_1, g_2\}$.

Notice that because $g_1 * g_2 = g_2 * g_1$, we can swap any two adjacent terms of c .

Therefore, we can rewrite c as

$$c = a * b = b_1 * b_2 * \cdots * b_q * a_1 * a_2 * \cdots * a_p. \quad (6)$$

Therefore, G is generated by g_1 and g_2 then G is abelian if and only if $g_1 * g_2 = g_2 * g_1$. $\square$

- (b) Suppose G is generated by $g_1, g_2, \dots, g_n$. Do you think that G is abelian if and only if $g_i * g_j = g_j * g_i$ for all $i, j \in \{1, 2, \dots, n\}$? Explain your reasoning, but you do not have to provide a proof.

I do think so. We can likely apply some inductive proof to show that adding a new g_k that is mutually commutative with all other g_i does not impact the ‘abelian-ness’ of the overall group G .

- (c) Prove that cyclic groups C_n are abelian.

Proof. By definition,

$$C_n = \langle r \mid r^n = e \rangle.$$

Because the cyclic groups C_n are defined by a generator with one non-identity element, then C_n must be abelian by associativity.

$$r^k * r^l = \underbrace{(r * r * \cdots * r)}_k * \underbrace{(r * r * \cdots * r)}_l \quad (7)$$

$$= \underbrace{(r * r * \cdots * r)}_l * \underbrace{(r * r * \cdots * r)}_k \quad (8)$$

$$= r^l * r^k. \quad (9)$$

$\square$

- (d) Prove that the general linear group of 2×2 real-valued matrices

$$GL_2(\mathbb{R}) = \{A \in \mathbb{R}^{n \times n} \mid \det(A) \neq 0\}$$

is not abelian.

Proof. Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, B = \begin{bmatrix} p & q \\ r & s \end{bmatrix}.$$

$$\text{Then, } A \times B = \begin{bmatrix} ap + br & aq + bs \\ cp + dr & cq + ds \end{bmatrix} \text{ and } B \times A = \begin{bmatrix} ap + cq & bp + dq \\ ar + cs & br + ds \end{bmatrix}.$$

In general, these are not the same matrix. $\square$

- (e) Prove that the dihedral group D_n is abelian if and only if $n \leq 2$.

Proof. First, by definition, the smallest dihedral group is D_1 , which is just the identity. This trivial group must be abelian.

Then, for $n = 2$,

$$D_2 = \langle r, f \mid r^2 = f^2 = e \rangle.$$

Notice that in this case, $r = f$, therefore, D_2 is abelian.

Then, to prove the “other direction”, we must prove that a dihedral group D_n being abelian implies that $n \leq 2$. We can prove this via contrapositive. Thus, we for a dihedral group D_n where $n > 2$, D_n is not abelian.

Suppose we have D_n where $n > 2$. We choose the element $rf \in D_n$. Now, $rf = fr^{n-1} \neq fr$, so D_n is not abelian.

Therefore, D_n is abelian if and only if $n \leq 2$. $\square$

3. The direct product of groups is a way of combining two groups into a new group. The notation can look a bit intimidating, but the basic idea is simple: just treat the groups independently. Here’s the formal definition, but keep in mind that $*$, $\star$, and $\odot$ are just names for binary operations like $+$, $\times$, and $\circ$.

Given two groups $(G_1, *)$ and $(G_2, \star)$, we define the direct product of groups as

$$G_1 \times G_2 = \{(g_1, g_2) \mid g_1 \in G_1 \text{ and } g_2 \in G_2\}$$

with the binary operation, $\odot$, where

$$(g_1, g_2) \odot (g'_1, g'_2) = (g_1 * g'_1, g_2 \star g'_2).$$

- (a) Let $(G_1, *) = (D_4, *)$ be the dihedral group of the square, and let $(G_2, \star) = (S_3, \circ)$ be the symmetric group under function composition. Show that in $G_1 \times G_2$,

$$(f, (132)) \odot (r, (23)) = (r^3 f, (13)).$$

$$(f, (132)) \odot (r, (23)) = (fr, (132) \circ (23)) \tag{10}$$

$$= (r^3 f, (2)(31)) = (r^3 f, (13)). \tag{11}$$

- (b) Prove that for any two groups G_1 and G_2 , the direct product of groups is a group.

Proof. Let $G = G_1 \times G_2$. We verify that G follows the group axioms:

Closure: Let $g_1, g'_1 \in G_1$ and $g_2, g'_2 \in G_2$. since $g_1 g'_1 \in G_1$, and $g_2 g'_2 \in G_2$, $(g_1 g'_1, g_2 g'_2) \in G$.

Identity: Let e_1, e_2 be the identity elements of G_1 and G_2 respectively. (e_1, e_2) is the identity of G .

Associativity: Since the operations on the first and second elements of G are both associative, associativity is inherited to the direct product.

Inverse: Similarly, because the left and right elements each independently have an inverse, the tuple containing both will be the equivalent inverse in G . $\square$

- (c) Prove that if $G_1 \times G_2$ is abelian if and only if G_1 and G_2 are both abelian.

First, if $G = G_1 \times G_2$ is abelian, then for $(a, b), (a', b') \in G$,

$$(a, b) \odot (a', b') = (a', b') \odot (a, b) \tag{12}$$

$$(a * a', b \star b') = (a' * a, b' \star b) \tag{13}$$

Since $a * a' = a' * a$ and $b \star b' = b' \star b$, G_1 and G_2 are abelian.

Now, given G_1 and G_2 are abelian, reverse the above algebra to see that G must also be abelian.

Thus, $G_1 \times G_2$ is abelian if and only if G_1 and G_2 are both abelian.

- (d) Let $\mathbb{Z}/k\mathbb{Z}$ denote the group of the integers mod k under addition. Let $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. Compute the order of G . Prove that G has an element of order $|G|$.

The order of a direct product of two groups is the product of the orders. Thus,

$$|G| = |\mathbb{Z}/2\mathbb{Z}| \times |\mathbb{Z}/3\mathbb{Z}| = 2 \times 3 = 6.$$

The element with order 6 is $(1, 1)$.

- (e) Let $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. Compute the order of G . Prove the G has no element of order $|G|$.

Similarly, $|G| = 2 \times 2 = 4$. Now, the elements of G are $\{(0, 0), (0, 1), (1, 0), (1, 1)\}$. The orders of all of these elements is 2.

Exercise 3 [1]

My favorite resource from this class is OpenSCAD. Although I will likely never use it again, it was very fun to play around with. The backstory for the development of it is also really cool.

Choice Exercise 4 [5]

1. [5] We sometimes describe groups by their presentation, $\langle S \mid R \rangle$, where S is a set of generators, and R is a set of relations. Recall, we define the dihedral group of the n -gon as

$$D_n = \langle \underbrace{r, f}_S \mid \underbrace{r^n = f^2 = (rf)^2 = e}_R \rangle.$$

What this means is that every element of $s \in D_n$ can be written as a word

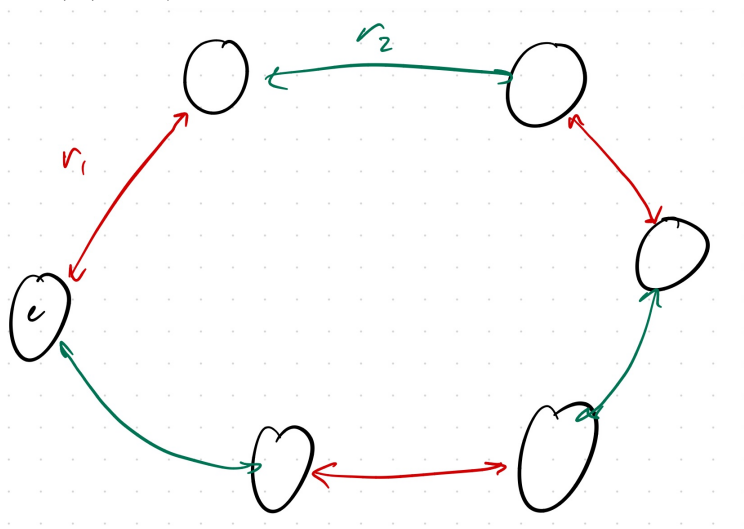
$$s = g_1^{e_1} g_2^{e_2} \dots g_m^{e_m},$$

where $g_i \in S = \{r, f\}$, $e_i \in \mathbb{Z}$, and where two words are equal if we can modify one to the other using the relations in R .

- (a) Draw a Cayley diagram of the group with presentation

$$G = \langle r_1, r_2 \mid r_1^2 = r_2^2 = (r_1 r_2)^3 = e \rangle.$$

(Hint: $|G| \leq 10$.)



- (b) Use the generators and relations to prove that $r^{-3} f r f^3 = r \in D_5$.

In D_5 , $r^5 = e$, $r^k f = f r^{n-k}$, $f^2 = e$, and $r^k = r^{5-k}$

Thus,

$$r^{-3} f r f^3 = r^2 f r f, \quad (14)$$

$$= r^2 f f r^4 \quad (15)$$

$$= r^6 = r. \quad (16)$$

- (c) Prove that for all positive integers k , $(r^k f)^2 = e \in D_n$.

Proof. We begin by expanding the express to $r^k f r^k f$. Then, we rewrite as $f r^{n-k} r^k f$. Notice that $r^{n-k} r^k = r^n = e$. Thus, we simplify down to $f f = e$. $\square$

- (d) Write a group presentation of the cyclic group C_n .

The group presentation of C_n is

$$C_n = \langle r \mid r^n = e \rangle.$$

Choice Exercise 6 [10]

- [3] Write a function that takes a word in S_n and outputs it in cycle notation. For instance, if the word is $\pi = 156243 \in S_n$ then the input to your program would be [1,5,6,2,4,3], and since $\pi = (254)(36)$ in cycle notation, the output would be [[254],[36]] or something equivalent.

```
word = [0,1,5,6,2,4,3] # prepend with a zero for convenience.
```

```
def word_to_cycle(word):
    explored = set()
    def explore(i):
        out = []
        while i not in explored:
            explored.add(i)
            out.append(i)
            i = word[i]
        return out

    out = [explore(i) for i in word]
    return list(filter(lambda x: len(x) > 1, out))
```

```
print(word_to_cycle(word)) # [[5, 4, 2], [6, 3]]
```

I was unfortunately not able to write this in Scheme LISP. I was defeated, but I managed to solve the other way

- [2] Write a function that goes the other way: it takes in input of a permutation in cycle notation (e.g. [[254],[36]]) and outputs the corresponding permutation as a word (e.g. [1,5,6,2,4,3].)

```
(define (max-vec v)
  (define (max-iter v champ i)
    (if (= i (vector-length v))
        champ
        (if (> (vector-ref v i) champ)
            (max-iter v (vector-ref v i) (+ i 1))
            (max-iter v champ (+ i 1))))))

(max-iter v (vector-ref v 0) 0))

(define (max-vec-vec vv)
  (define (max-iter vv champ i)
    (if (= i (vector-length vv))
        champ
        (let ((m (max-vec (vector-ref vv i))))
          (if (> m champ)
              (max-iter vv m (+ i 1))
              (max-iter vv champ (+ i 1))))))
  (if (= (vector-length vv) 0)
      (error "Empty vector")
      (max-iter vv (max-vec (vector-ref vv 0)) 1)))

(define (cycle-to-word cycles)
  (define m (+ 1 (max-vec-vec cycles)))
  (define word (make-vector m 0))
  (define (cycle-iter cycle i)
```

```

(define n (vector-length cycle))
(if (< i n)
    (begin
      (vector-set!
        word
        (vector-ref cycle i)
        (vector-ref cycle (modulo (+ 1 i) n)))
      (cycle-iter cycle (+ 1 i))
    ))
(define l (vector-length cycles))
(define (cycles-iter i)
  (if (< i l)
      (begin
        (cycle-iter (vector-ref cycles i) 0)
        (cycles-iter (+ 1 i))
      )
      )
  )
(cycles-iter 0)
word
)

```

3. [2] Write a function that takes two permutations as words and composes them, working right-to-left as usual.

```

word1 = [0,6,2,1,4,3,5]
print(word, word1)

def compose_word(a, b):
    out = [0] * max(len(a), len(b))

    for i in range(1, len(out)):
        out[i] = b[a[i]] if a[i] < len(b) else a[i]

    return out

print(compose_word(word, word1)) # [0, 6, 3, 5, 2, 4, 1]

```

4. [2] Write a function that takes two permutations in cycle notation and composes them, working right-to-left as usual.

```

def cycle_to_word(cycles):
    n = 1 + max(max(cycle) for cycle in cycles)
    word = list(range(n))

    for cycle in cycles:
        for x, y in zip(cycle, cycle[1:] + [cycle[0]]):
            word[x] = y

    return word

print(cycle_to_word(word_to_cycle(word))) # [0,1,5,6,2,4,3]

def compose_cycle(a, b):
    x = cycle_to_word(a)

```

```

    y = cycle_to_word(b)
    return word_to_cycle(compose_word(x, y))

p1 = [[2,5,4],[3,6]]
p2 = [[4,5],[1,2]]
print(compose_cycle(p1, p2)) #[0, 2, 4, 6, 1, 5, 3]

```

5. [1] Write two functions that compute the order of a permutation, one that works on words, and one that works on cycle notation.

```

def order_cycle(cycles):
    curr = cycles
    i = 1

    while len(curr) > 0:
        i += 1
        curr = compose_cycle(cycles, curr)

    return i

def order_word(word):
    curr = word
    i = 1
    while not all(index == x for (index, x) in enumerate(curr)):
        i += 1
        curr = compose_word(word, curr)

    return i

print(order_cycle(p1))
print(order_word(cycle_to_word(p1)))

```